NAME	Sheheryar Shakeel
ROLL NO	26K-3098

BANK MANAGEMENT ASSIGNMENT

SOURCE CODE:

bank_management.c

```c
#include <stdio.h>
int calculateTotal(int *arr, int size){
    int total = 0;
    int i;
    for (i = 0; i < size; i++)
    {
        total += *(arr + i);
    }
    return total;
}
int deposit(int a, int b){
    return a + b;
}
int withdraw(int a, int b){
    return a - b;
}
int processTransaction(int a, int b, int (*op)(int, int)){
    return op(a, b);
}
int main(){
    int i, j;
    int balance[5] = {15000, 25000, 18000, 32000, 12000};
    int *ptr;
    int choice;

    printf("Task 1\n");
    printf("Enter customer number (1-5): ");
    scanf("%d", &choice);
    ptr = &balance[choice - 1];
    printf("Balance: %d\n", *ptr);
    printf("Address: %p\n", (void *)ptr);
```

C bank_management.c

```c
int main(){
    printf("Address: %p\n", (void *)ptr);
    *ptr = *ptr + 5000;
    printf("Updated Balance: %d\n", *ptr);

    printf("\nTask 2\n");
    int transactions[2][3] = {{120000, 150000, 180000}, {200000, 170000, 220000}};
    int (*rowPtr)[3] = transactions;
    int sum;

    for (i = 0; i < 2; i++){
        sum = 0;
        printf("Branch %d Row Address: %p\n", i + 1, (void *)(rowPtr + i));
        for (j = 0; j < 3; j++){
            printf("Month %d: %d\n", j + 1, *(*(rowPtr + i) + j));
            sum += *(*(rowPtr + i) + j);
        }
        printf("Total Transactions of Branch %d: %d\n", i + 1, sum);
    }

    printf("\nTask 3\n");
    int branch1[3] = {15000, 23000, 32000};
    int branch2[4] = {12000, 21000, 27000, 35000};
    int *branches[2] = {branch1, branch2};
    int sizes[2] = {3, 4};
    int total;

    for (i = 0; i < 2; i++){
        total = 0;
        printf("Branch %d\n", i + 1);
        for (j = 0; j < sizes[i]; j++){
```

C bank_management.c

```c
int main(){
    for (i = 0; i < 2; i++){
        for (j = 0; j < sizes[i]; j++){
            printf("Balance: %d Address: %p\n", *(*(branches + i) + j), (void *)(*(branches + i) + j));
            total += *(*(branches + i) + j);
        }
        printf("Total Balance of Branch %d: %d\n", i + 1, total);
    }

    printf("\nTask 4\n");
    printf("Branch 1 Total: %d\n", calculateTotal(branch1, 3));
    printf("Branch 2 Total: %d\n", calculateTotal(branch2, 4));
    printf("\nTask 5\n");
    int (*fptr)(int, int);
    int amount = 2000;
    int current = 10000;
    fptr = deposit;
    printf("After Deposit: %d\n", fptr(current, amount));

    fptr = withdraw;
    printf("After Withdraw: %d\n", fptr(current, amount));
    printf("\nTask 6\n");
    printf("Deposit Result: %d\n", processTransaction(current, amount, deposit));
    printf("Withdraw Result: %d\n", processTransaction(current, amount, withdraw));
    return 0;
}
```

OUTPUT:

```
Task 1
Enter customer number (1-5): 2
Balance: 25000
Address: 0061FEEC
Updated Balance: 30000

Task 2
Branch 1 Row Address: 0061FECC
Month 1: 120000
Month 2: 150000
Month 3: 180000
Total Transactions of Branch 1: 450000
Branch 2 Row Address: 0061FED8
Month 1: 200000
Month 2: 170000
Month 3: 220000
Total Transactions of Branch 2: 590000

Task 3
Branch 1
Balance: 15000 Address: 0061FEC0
Balance: 23000 Address: 0061FEC4
Balance: 32000 Address: 0061FEC8
Total Balance of Branch 1: 70000
Branch 2
Balance: 12000 Address: 0061FEB0
Balance: 21000 Address: 0061FEB4
```

```
Balance: 27000 Address: 0061FEB8
Balance: 35000 Address: 0061FEBC
Total Balance of Branch 2: 95000

Task 4
Branch 1 Total: 70000
Branch 2 Total: 95000

Task 5
After Deposit: 12000
After Withdraw: 8000

Task 6
Deposit Result: 12000
Withdraw Result: 8000
PS C:\Users\I C\VS\folderofc.c>
```